10TH CLASS CHEMISTRY — NEW SYLLABUS 2025-26 (PECTAA)

Chapter 15 — Stoichiometry

Solved Quick Checks + Exercise | New Book (PECTAA 2025-26)

Quick Checks (Solved)

Quick Check 15.1

1. Calculate the amount of CO_2 in 100 dm^3 at RTP.

At RTP one mole of every gas occupies 24 dm^3, so the number of moles is found by dividing the volume by 24.

Number of moles = Volume ÷ Molar volume = 100 dm^3 ÷ 24 dm^3 mol^{-1} = 4.17 mol

Mass of CO_2 = moles × molar mass = 4.17 mol × 44 g mol^{-1} = 183.5 g

So 100 dm^3 of CO_2 at RTP contain 4.17 moles, which is 183.5 g of carbon dioxide.

Quick Check 15.2

1. What mass of $CuSO_4$ is present in 50 cm^3 of a 5 × 10^{-2} mol dm^{-3} aqueous solution?

First change the volume into dm^3, then find the moles by multiplying the molar concentration with the volume, and finally change the moles into mass.

Volume = 50 cm^3 ÷ 1000 = 0.05 dm^3

Moles of $CuSO_4$ = concentration × volume = 5 × 10^{-2} mol dm^{-3} × 0.05 dm^3 = 2.5 × 10^{-3} mol

Molar mass of $CuSO_4$ = 64 + 32 + (4 × 16) = 160 g mol^{-1}

Mass = moles × molar mass = 2.5 × 10^{-3} mol × 160 g mol^{-1} = 0.40 g

So 0.40 g of copper sulphate is present in the solution.

Quick Check 15.3

1. Calculate the volume of 0.1 mol dm^{-3} oxalic acid ($C_2O_4H_2$) solution needed to exactly neutralize 25 cm^3 of 0.1 mol dm^{-3} KOH solution.

Write the balanced equation first, because the mole ratio of acid and base is needed.

$$H_2C_2O_4(aq) + 2KOH(aq) \rightarrow K_2C_2O_4(aq) + 2H_2O(l)$$

Volume of KOH = 25 cm^3 = 0.025 dm^3

Moles of KOH = 0.1 mol dm^{-3} × 0.025 dm^3 = 2.5 × 10^{-3} mol

From the equation, 2 moles of KOH neutralize 1 mole of oxalic acid, so

Moles of oxalic acid = 2.5 × 10^{-3} ÷ 2 = 1.25 × 10^{-3} mol

Volume of acid = moles ÷ concentration = 1.25 × 10^{-3} mol ÷ 0.1 mol dm^{-3} = 0.0125 dm^3

Volume of oxalic acid = 0.0125 × 1000 = 12.5 cm^3

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Quick Check 15.4

1. Calculate the percentage of sulphate ions (SO_4^{2-}) in $Al_2(SO_4)_3$.

Formula mass of $Al_2(SO_4)_3$ = (2 × 27) + 3 × [32 + (4 × 16)] = 54 + 288 = 342

Mass of sulphate ions in one formula unit = 3 × 96 = 288

$$\text{Percentage of sulphate} = \frac{288}{342} \times 100 = 84.2\ \%$$

So 84.2 % of the mass of aluminium sulphate is due to the sulphate ions.

Quick Check 15.5

1. A 0.5 g sample of a compound contains 0.418 g of antimony (Sb) and 0.082 g of oxygen. What is the empirical formula of the compound?

Step 1: find the moles of each element by dividing its mass by its atomic mass.

Moles of Sb = 0.418 g ÷ 121.8 g mol^{-1} = 3.43 × 10^{-3} mol

Moles of O = 0.082 g ÷ 16 g mol^{-1} = 5.13 × 10^{-3} mol

Step 2: divide both values by the smallest number of moles.

Sb = 3.43 ÷ 3.43 = 1.00 O = 5.13 ÷ 3.43 = 1.50

Step 3: the ratio 1 : 1.5 is not a whole number ratio, so multiply it by 2.

Sb = 2, O = 3. Hence the empirical formula of the compound is **Sb_2O_3**.

Quick Check 15.6

1. An unknown hydrocarbon contains 85.71 % carbon. Its molar mass is 84 g mol^{-1}. What is its molecular formula?

Percentage of hydrogen = 100 − 85.71 = 14.29 %

Moles of C = 85.71 ÷ 12 = 7.14 Moles of H = 14.29 ÷ 1 = 14.29

Atomic ratio C : H = 7.14 : 14.29 = 1 : 2, so the empirical formula is CH and its formula mass is 12 + 1 = 13.

n = molar mass ÷ empirical formula mass = 84 ÷ 13 ≈ 6

Molecular formula = n × empirical formula = 6 × CH = **C_6H_{12}** (hexene).

Quick Check 15.7

1. Why is it necessary for a chemical equation to be balanced before it can be used in calculation?

A balanced chemical equation gives the correct mole ratio in which the reactants combine and the products are formed, for example 1 mole of $CaCO_3$ gives 1 mole of $CaCl_2$. All stoichiometric calculations depend on this ratio. If the equation is not balanced, the mole ratio is wrong, so every calculated mass, volume or number of moles of the product will be wrong. Balancing also satisfies the law of conservation of mass, since the number of atoms of each element must be the same on both sides.

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Quick Check 15.8

1. If you react 100 g of chlorine gas with 500 g of KI, which reactant will be the limiting reactant? (Cl_2 + 2KI → 2KCl + I_2)

Find the number of moles of each reactant and then see how much product each can give.

Moles of Cl_2 = 100 g ÷ 71 g mol^{-1} = 1.41 mol

From the equation, 1 mole of Cl_2 gives 1 mole of I_2, so Cl_2 can give 1.41 mol of I_2.

Moles of KI = 500 g ÷ 166 g mol^{-1} = 3.01 mol

From the equation, 2 moles of KI give 1 mole of I_2, so KI can give 3.01 ÷ 2 = 1.51 mol of I_2.

Chlorine gives the smaller amount of product (1.41 mol), so **chlorine is the limiting reactant** and potassium iodide is present in excess.

Quick Check 15.9

1. Find the theoretical yield and percent yield of oxygen by heating 40 g of $KClO_3$ (M = 122.5). The mass of oxygen produced is 14.9 g. ($2KClO_3$ → 2KCl + $3O_2$)

Moles of $KClO_3$ = 40 g ÷ 122.5 g mol^{-1} = 0.3265 mol

From the equation, 2 moles of $KClO_3$ give 3 moles of O_2, so

Moles of O_2 = 0.3265 × 3 ÷ 2 = 0.4898 mol

Theoretical yield of oxygen = 0.4898 mol × 32 g mol^{-1} = 15.67 g

$$\text{Percentage yield} = \frac{14.9}{15.67} \times 100 = 95.1\ \%$$

So the theoretical yield of oxygen is 15.67 g and the percentage yield is 95.1 %.

Exercise (Solved)

A. Multiple Choice Questions

1. Volume occupied by 15 moles of ammonia gas at RTP:

(a) 224 dm^3 (b) 160 dm^3

(c) 360 dm^3 (d) 265 dm^3

Answer: (c) — Volume = moles × 24 = 15 × 24 = 360 dm^3.

2. How many moles of natural gas CH_4 will be present in a cylinder if its volume is 18 dm^3 at RTP?

(a) 0.75 mol (b) 0.67 mol

(c) 0.85 mol (d) 0.56 mol

Answer: (a) — Moles = volume ÷ molar volume = 18 ÷ 24 = 0.75 mol.

3. What will be the molar concentration of potassium hydroxide solution if 2.0 g of it are dissolved in 100 cm^3 of solution?

(a) 0.36 mol dm^{-3} (b) 0.53 mol dm^{-3}
(c) 0.70 mol dm^{-3} (d) 0.45 mol dm^{-3}

Answer: (a) — Moles = 2.0 ÷ 56 = 0.0357 mol; volume = 0.1 dm^3; concentration = 0.0357 ÷ 0.1 = 0.36 mol dm^{-3}.

4. What will be the mass of NaCl needed to make up 15 dm^3 of its solution with a concentration of 0.7 g dm^3?

(a) 21 g (b) 10.5 g
(c) 31.5 g (d) 5.4 g

Answer: (b) — Mass = concentration in g dm^{-3} × volume = 0.7 × 15 = 10.5 g.

5. Find out the concentration of H_2SO_4 in mol dm^{-3} if its 10 cm^3 react with 20 cm^3 of NaOH solution having concentration 0.1 mol dm^{-3}.

(a) 0.01 mol dm^{-3} (b) 0.001 mol dm^{-3}
(c) 0.1 mol dm^{-3} (d) 0.15 mol dm^{-3}

Answer: (c) — $H_2SO_4 + 2NaOH \rightarrow Na_2SO_4 + 2H_2O$; moles of NaOH = 0.1 × 0.020 = 0.002, so moles of acid = 0.001 in 0.010 dm^3, that is 0.1 mol dm^{-3}.

6. If the molar mass of a compound is 178 g mol^{-1} and its empirical formula is $C_3H_5O_3$, what is its molecular formula?

(a) $C_3H_5O_3$ (b) $C_6H_{10}O_6$
(c) $C_9H_{15}O_9$ (d) $C_8H_{10}O_8$

Answer: (b) — Empirical formula mass = 36 + 5 + 48 = 89; n = 178 ÷ 89 = 2, so the molecular formula is $C_6H_{10}O_6$.

7. When natural gas burns in air, which component is the limiting reactant?

(a) Both components (b) Neither component is a limiting reactant
(c) Air (d) Natural gas

Answer: (d) — Air is taken in large excess, so natural gas is used up first and controls the amount of product.

8. If the actual yield of a compound is 0.198 g while the theoretical yield is 0.217 g, what will be the percentage yield?

(a) 50 % (b) 91.2 %
(c) 80 % (d) 90 %

Answer: (b) — Percentage yield = (0.198 ÷ 0.217) × 100 = 91.2 %.

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B. Short Answer Questions

15.1 What is the molar volume of a gas at RTP?

The volume occupied by one mole of any gas at room temperature and pressure is called its molar volume. At RTP (25 °C or 298.15 K and 1 atmosphere or 760 torr) one mole of every gas occupies 24 dm^3. Its value is therefore 24 dm^3 mol^{-1}. The masses and sizes of the molecules of a gas do not change this volume.

15.2 How does the concept of molar volume useful for us?

The concept of molar volume is useful because it connects the volume of a gas with its number of moles. If the volume of a gas at RTP is known, its moles are found by dividing the volume by 24, and then its mass can be found by multiplying the moles with the molar mass. In the same way, if the mass or the moles of a gas are known, its volume at RTP can be calculated. This is very helpful because gases are usually measured by volume and not by mass.

15.3 How does the molar volume relate to the Avogadro's law?

Avogadro's law states that equal volumes of all gases contain an equal number of molecules at the same temperature and pressure. One mole of every substance contains the same number of particles (6.02×10^{23}). Therefore one mole of every gas contains the same number of molecules and, according to Avogadro's law, it must occupy the same volume at the same temperature and pressure. This common volume is the molar volume, which is 24 dm^3 at RTP.

15.4 Why two different compounds may show the same empirical formula?

The empirical formula of a compound shows only the simplest whole number ratio of the atoms present in it; it does not tell the actual number of atoms in one molecule. Different compounds may have the same simplest ratio, and then they have the same empirical formula although their molecular formulae are different. For example, acetylene (C_2H_2) and benzene (C_6H_6) both have the empirical formula CH, and acetic acid (CH_3COOH) and glucose ($C_6H_{12}O_6$) both have the empirical formula CH_2O.

15.5 Why is percentage yield important?

Percentage yield tells us how efficient a reaction is; it compares the actual yield obtained in the laboratory or factory with the theoretical yield calculated from the balanced equation. A high percentage yield means that most of the reactants have been converted into the product, which saves raw material, time and money. In industry and in the preparation of medicines, it helps the chemists to judge and improve the method, to reduce waste and to keep the cost of the product low.

15.6 What is the concentration of a solution in mol dm^{-3} if it contains 49 g of H_2SO_4 in one dm^3 of solution?

Molar mass of H_2SO_4 = (2 × 1) + 32 + (4 × 16) = 98 g mol^{-1}

Moles of H_2SO_4 = 49 g ÷ 98 g mol^{-1} = 0.5 mol

Volume of solution = 1 dm^3, so concentration = 0.5 mol ÷ 1 dm^3 = **0.5 mol dm^{-3}**

C. Constructed Response Questions

15.1 How would you identify the limiting reactant in a chemical reaction?

The limiting reactant is that reactant which is consumed completely first and which controls the amount of the product formed. It is identified in the following steps.

Step 1: Calculate the number of moles of each reactant from the given masses or volumes by using its molar mass or molar volume.

Step 2: Using the balanced chemical equation, calculate the number of moles of the product which each reactant can give.

Step 3: The reactant which gives the least amount of the product is the limiting reactant; the other reactant is present in excess and is left behind after the reaction.

For example, in $3H_2 + N_2 \rightarrow 2NH_3$, 12 g of H_2 (6 mol) can give 4 mol of NH_3 while 64 g of N_2 (2.29 mol) can give 4.58 mol of NH_3. Hydrogen gives the smaller amount, so hydrogen is the limiting reactant.

15.2 Why is it important to find the purity of a compound which is used as a medicine?

The dose written on a medicine is correct only if the medicine contains the required amount of the pure active drug. If the medicine is impure, its actual dose becomes less than the required dose, so it may be weak and useless, or the impurities present in it may cause harmful side effects in the patient. Percentage purity is found by dividing the mass of the pure drug by the total mass of the sample and multiplying by 100. This test makes sure that the patient gets the correct dose, so it keeps the treatment safe and effective.

15.3 Will there be a limiting reactant in a reversible reaction?

No, the idea of a limiting reactant does not apply to a reversible reaction. A reversible reaction does not go to completion; it reaches a state of equilibrium in which the forward and the reverse reactions take place at the same rate. At equilibrium both the reactants and the products are present together, and no reactant is used up completely. Since no reactant finishes first, there is no limiting reactant and the yield of such a reaction is less than the theoretical yield.

15.4 How do you know if a formula is empirical or not?

Look at the subscripts of the formula and check whether they are in the simplest whole number ratio. If the subscripts cannot be divided by any common number, the formula is already an empirical formula, for example H_2O, NaCl and $MgCl_2$. If the subscripts can still be divided by a common number, the formula is a molecular formula and not an empirical formula; for example C_2H_4 can be divided by 2 to give CH_2, so C_2H_4 is a molecular formula. All ionic compounds are represented by their empirical formulae because their formulae already show the simplest ratio of the ions.

15.5 Write down one method to find out the molecular mass of a compound without knowing its molecular formula.

Molecular mass can be found by dividing the mass of a known number of moles of the compound by that number of moles. A known mass of the compound is taken and the number of moles in it is found; for a gas, its volume at RTP is measured and the moles are found by dividing the volume by 24. Then molecular mass = mass ÷ moles.

For example, if 0.320 mol of a compound weighs 28.8 g, its molecular mass = 28.8 g ÷ 0.320 mol = 90 g mol^{-1}. This value can then be used with the empirical formula mass to find the molecular formula of the compound.

D. Numerical Problems

15.1 Calculate the number of moles, volume and number of molecules in 4.0 g of CH_4 at RTP.

Molar mass of CH_4 = 12 + (4 × 1) = 16 g mol^{-1}

Moles = 4.0 g ÷ 16 g mol^{-1} = 0.25 mol

Volume = moles × molar volume = 0.25 × 24 = 6.0 dm^3

Number of molecules = moles × 6.02 × 10^{23} = 0.25 × 6.02 × 10^{23} = 1.505 × 10^{23} molecules

So 4.0 g of methane contains 0.25 mol, occupies 6.0 dm^3 at RTP and has 1.505 × 10^{23} molecules.

15.2 What mass of NaCl needs to be dissolved in 150 cm^3 of a solution if the concentration of the solution is 0.4 mol dm^{-3}?

Volume = 150 cm^3 ÷ 1000 = 0.15 dm^3

Moles of NaCl = concentration × volume = 0.4 × 0.15 = 0.06 mol

Molar mass of NaCl = 23 + 35.5 = 58.5 g mol^{-1}

Mass = moles × molar mass = 0.06 × 58.5 = 3.51 g

So 3.51 g of sodium chloride is needed.

15.3 Find out the percentage of sodium in $NaHCO_3$.

Formula mass of $NaHCO_3$ = 23 + 1 + 12 + (3 × 16) = 84

$$\text{Percentage of Na} = \frac{23}{84} \times 100 = 27.4\ \%$$

So sodium makes 27.4 % of the mass of sodium bicarbonate.

15.4 A sulphide of iron contains 1.926 g of sulphur and 2.333 g of iron. Find out the empirical formula of this compound.

Moles of Fe = 2.333 g ÷ 55.85 g mol^{-1} = 0.0418 mol

Moles of S = 1.926 g ÷ 32.06 g mol^{-1} = 0.0601 mol

Atomic ratio Fe : S = 0.0418 : 0.0601 = 1 : 1.44

The ratio is not a whole number, so multiply by 2: Fe = 2, S = 2.88 ≈ 3

Hence the empirical formula of the sulphide is **Fe_2S_3**.

15.5 A compound contains by mass 40.0 % C, 6.71 % H and 53.3 % oxygen. A 0.320 mole of this compound weighs 28.8 g. What is the molecular formula of this compound?

Moles of C = 40.0 ÷ 12 = 3.33 Moles of H = 6.71 ÷ 1 = 6.71 Moles of O = 53.3 ÷ 16 = 3.33

Atomic ratio C : H : O = 3.33 : 6.71 : 3.33 = 1 : 2 : 1, so the empirical formula is CH_2O and its formula mass is 12 + 2 + 16 = 30.

Molecular mass = mass ÷ moles = 28.8 g ÷ 0.320 mol = 90 g mol^{-1}

n = molecular mass ÷ empirical formula mass = 90 ÷ 30 = 3

Molecular formula = 3 × CH_2O = $\mathbf{C_3H_6O_3}$

15.6 The formation of ammonia gas is given by the following equation: $\mathbf{3H_{2(g)} + N_{2(g)} \rightarrow 2NH_{3(g)}}$. If you start reacting 12 g of $\mathbf{H_2}$ with 64 g of $\mathbf{N_2}$, which will be the limiting reactant?

Moles of H_2 = 12 g ÷ 2 g mol^{-1} = 6 mol

From the equation, 3 moles of H_2 give 2 moles of NH_3, so 6 mol of H_2 give 6 × 2 ÷ 3 = 4 mol of NH_3.

Moles of N_2 = 64 g ÷ 28 g mol^{-1} = 2.29 mol

From the equation, 1 mole of N_2 gives 2 moles of NH_3, so 2.29 mol of N_2 give 2.29 × 2 = 4.57 mol of NH_3.

Hydrogen gives the smaller amount of ammonia (4 mol), so **hydrogen is the limiting reactant** and nitrogen is left in excess.

15.7 When 305 g of $\mathbf{AgNO_3}$ is reacted with excess of $\mathbf{MgCl_2}$, it produces 23.7 g of $\mathbf{Mg(NO_3)_2}$. What is the percentage yield of the reaction?
$\mathbf{2AgNO_{3(aq)} + MgCl_{2(aq)} \rightarrow Mg(NO_3)_{2(aq)} + 2AgCl(s)}$

Molar mass of $AgNO_3$ = 108 + 14 + (3 × 16) = 170 g mol^{-1}

Moles of $AgNO_3$ = 305 g ÷ 170 g mol^{-1} = 1.794 mol

From the equation, 2 moles of $AgNO_3$ give 1 mole of $Mg(NO_3)_2$, so moles of $Mg(NO_3)_2$ = 1.794 ÷ 2 = 0.897 mol

Molar mass of $Mg(NO_3)_2$ = 24 + 2 × (14 + 48) = 148 g mol^{-1}

Theoretical yield = 0.897 mol × 148 g mol^{-1} = 132.8 g

$$\text{Percentage yield} = \frac{23.7}{132.8} \times 100 = 17.8\ \%$$

So the percentage yield of magnesium nitrate is only 17.8 %.

15.8 A sample of metal has a total mass of 3.66 kg and contains 2.45 kg of gold. What is the percentage purity of gold in this sample?

$$\text{Percentage purity} = \frac{2.45\ \text{kg}}{3.66\ \text{kg}} \times 100 = 66.9\ \%$$

So the percentage purity of gold in the sample is 66.9 %; the rest of the sample is made of other metals or impurities.